AeroMorse Switch Mode Guide — Two-Button Hold Mode

Switch mode (Group 9; new in v1.15 as Group 7, moved to Group 9 in v1.24) turns AeroMorse into **two plain switches**. There is no Morse code in this group:

- a **sip** presses **Enter** the moment it starts and **holds it** until the sip ends;
- a **puff** presses **Space** the moment it starts and **holds it** until the puff ends.

That is exactly how a pair of physical accessibility switches behaves, so it works with games and apps built for one or two switches — including ones that need a key **held down**, which Morse groups can't do.

When to use which group

	Group 4 — Scanning	**Group 9 — Switch**
What a sip/puff does	Taps a key after the Morse pattern finishes	Presses a key **instantly** and **holds** it while you sip/puff
Keys	12: Enter, Space, F3–F12 (1–3-symbol patterns)	2: Enter (sip), Space (puff)
Holding a key down	No	**Yes**
Best for	iOS Switch Control, Android Switch Access, Samsung Universal Switch	Switch games and apps, anything that needs "hold Space"
Guide	AEROMORSE_SWITCH_CONTROL_GUIDE.md	this guide

Quick start

1. **Enter Switch mode:** type the Group 0 code `. - - - - - - -` (1 sip, 7 puffs). The screen shows **[SWITCH]**. (Long-press group cycling skips Switch mode, so you never land in it by accident.)
2. **Sip** to hold **Enter**, **puff** to hold **Space**. While a key is held the screen shows `HOLD RETURN` or `HOLD SPACEBAR`.
3. **Leave Switch mode:** just **stop** — after **20 seconds** with no sip or puff it goes back to **Keyboard** by itself. (Or hold one puff for **5 seconds**: Space is let go at 5 s and you switch to Keyboard.)

Things that do **not** change group in Switch mode, on purpose:

- **Hard (strong) sips and puffs** — games often need them.
- **Long sips and puffs** — games often need long holds. (The one exception is a puff held for the full `SWITCH_EXIT_PUFF_S`, 5 seconds as supplied, which leaves Switch mode. A sip can be held as long as you like.)
- **Quick repeated sips or puffs.**

The first sip or puff after AeroMorse starts only closes the start-up (version) screen; it doesn't press a key.

Benny's Hub

Benny's Hub (by NARBE) is a free library of games and tools made to be used with **one or two buttons mapped to Space and Return** — a good match for Switch mode, and it runs in any web browser on the computer AeroMorse is plugged into.

How the hub is driven:

AeroMorse (Switch mode)	Key	In Benny's Hub
Puff	Space	Move to the next item
Sip	Enter (Return)	Choose / open the highlighted item

Games that need a held key. Some games use how long you hold. In **NARBE Kart** (the racing game), for example, the on-screen help says *Hold Space to slide left, hold Enter to slide right*, and its menus use *tap Space = next, hold Space = back, Enter = choose*. In Switch mode that is simply:

- **hold a puff** → hold Space → slide left;
- **hold a sip** → hold Enter → slide right;
- short puff → next; long puff → back; sip → choose.

Getting started:

1. On the computer, open https://narbehouse.github.io/bennyshub/index.html in a browser (full screen helps).
2. On AeroMorse, enter Switch mode (`.-------`).
3. Puff to move, sip to choose. When you're finished, stop for 20 seconds and AeroMorse returns to Keyboard.

Some Benny's Hub games also have a **one-switch** option. In Switch mode you can use just the puff (Space) for those.

How it compares with a two-switch adapter

NARBE also publish **Benny's Trinkey**, a small USB adapter that turns two plugged-in switches into Space and Enter (code on GitHub). Switch mode does the same job — each key goes down when the press starts and comes up when it ends — with these differences:

	Benny's Trinkey	AeroMorse Switch mode
Which input is which	Switch 1 = Space, switch 2 = Enter	Sip = Enter, puff = Space (change with `SWITCH_SIP_KEY` / `SWITCH_PUFF_KEY`)
Both keys held together	Yes	**No** — one at a time
Longest hold	No limit	Enter (sip): no limit. Space (puff): up to `SWITCH_EXIT_PUFF_S`, 5 seconds as supplied
Leaves by itself	Never — it is always Space and Enter	After `SWITCH_IDLE_EXIT_S` (20 seconds) with no sip or puff
Shows what is held	Light on the adapter	`HOLD SPACEBAR` / `HOLD RETURN` on the display

What that means in practice:

- **A game that needs Space and Enter held at the same time will not work.** You cannot sip and puff together. On a two-switch build, if both switches are pressed the dot switch (Enter) is the one that counts.
- **Going straight from a sip to a puff** (or back) lets go of the first key and then presses the other.
- **A game that needs Space held longer than 5 seconds** will drop you out of Switch mode part-way through. Raise `SWITCH_EXIT_PUFF_S`, for example to `10`. The 20-second idle exit still gets you out. Or swap the two keys so the long hold is on the sip, which has no limit.

Settings (`config.py`)

All in the **SWITCH GROUP** section of `config.py`. Save and the device restarts with the new values.

Setting	Default	What it does
`SWITCH_GROUP`	`9`	Which group is Switch mode (it was `7` before v1.24). `0` = no Switch mode (Group 9 becomes an ordinary group again).
`SWITCH_SIP_KEY`	`"ENTER"`	Key held while sipping. Any Keycode name, e.g. `"LEFT_ARROW"`.
`SWITCH_PUFF_KEY`	`"SPACE"`	Key held while puffing, e.g. `"RIGHT_ARROW"`.
`SWITCH_IDLE_EXIT_S`	`20`	Seconds with no sip/puff before it returns to `SWITCH_EXIT_GROUP` by itself. `0` = never (then use the long puff).
`SWITCH_EXIT_PUFF_S`	`5.0`	Hold one puff this long to leave straight away. This is also the longest the puff key can be held, so raise it (e.g. `10`) for games with long Space holds. Lowest allowed: `2`.
`SWITCH_EXIT_GROUP`	`1`	Where you go when leaving (1 = Keyboard).
`SWITCH_COUNTDOWN_S`	`10`	(v1.21+) Seconds of on-screen countdown before it leaves, e.g. `KEYBOARD IN 5` — it names the group you are going to. It runs before the idle exit, and during a long puff from 1 second into the hold. `0` = none.

Key names are listed in the **Keycode Reference** (`KEYCODE_REFERENCE.md` / the PDF). For example, a game controlled with the arrow keys could use `SWITCH_SIP_KEY = "LEFT_ARROW"` and `SWITCH_PUFF_KEY = "RIGHT_ARROW"`.

The validator (`Check my AeroMorse files`) checks these settings before you copy `config.py` to the device.

Tips

- **Watch the top line.** It says **[SWITCH]** while you're in Switch mode. A wireless display mirrors it, so you can put it where you can see it while playing.
- **Resting mid-game** for less than 20 seconds keeps you in Switch mode. If your game has longer pauses, raise `SWITCH_IDLE_EXIT_S` (e.g. `60`).
- **Keys never get stuck.** The key is let go when your sip/puff ends, and also whenever AeroMorse leaves Switch mode.
- **The sensor settings still apply** — `THRESH_SIP` / `THRESH_PUFF` decide how hard a sip/puff must be to count, the same as for Morse.

Troubleshooting

Problem	Why	Fix
It leaves Switch mode while I'm playing	A pause longer than `SWITCH_IDLE_EXIT_S`, or a 5-second puff	Raise `SWITCH_IDLE_EXIT_S` / `SWITCH_EXIT_PUFF_S`
Space lets go during a long hold	The puff reached `SWITCH_EXIT_PUFF_S` and left Switch mode	Raise `SWITCH_EXIT_PUFF_S` (e.g. `10`)
The game wants Space and Enter together	Switch mode holds one key at a time	Not possible with sip-and-puff; choose a game setting that uses one key at a time
I need the mouse between games	Switch mode has no Morse, so there is no code to press	Set `SWITCH_EXIT_GROUP = 2` so leaving lands in Mouse, and add a short Group 2 code for `"group 9"` in `morse_map.py` to come back. The countdown (`SWITCH_COUNTDOWN_S`) shows when it is about to leave
It won't leave Switch mode	You keep sipping/puffing within 20 s	Stop for 20 s, or hold one puff for 5 s
Nothing happens in the game	The game window isn't in front, or wants other keys	Click the game once; or change `SWITCH_SIP_KEY` / `SWITCH_PUFF_KEY`
`.-------` doesn't go to Switch mode	`SWITCH_GROUP` isn't `9` (before v1.24 it was `7`, reached with `...-----`)	Check `config.py` (or use the Group 0 code for the group you chose)
Group 9's own patterns don't type	Group 9 is Switch mode	Set `SWITCH_GROUP = 0` to use Group 9 for Morse again